Forgetting to Refuse: Does a Taught Safeguard Survive Benign Personalization of a Vision-Language-Action Policy?

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Abstract: Benign fine-tuning degrades refusal in aligned language models; whether an *action-level* safeguard in a vision-language-action (VLA) policy survives benign personalization is unmeasured. In two pre-registered studies we teach OpenVLA-7B to hold still under harmful instructions, personalize it on 200 benign demonstrations, and re-measure refusal and closed-loop contact with a static human hand against a comparator never taught the safeguard. Study 1's primary outcome, harmful-instruction contact, was null (0/50 before and after), but no arm showed evidence of executing the harmful clause, so it cannot register compliance; exploratory results showed the aligned policy losing its benign-instruction advantage (2% vs. the comparator's 72% contact before personalization, 16% vs. 10% after) and a first-step refusal rate that also rose in the untaught comparator. Study 2, on ten fresh scenes, confirmed the loss: benign-instruction contact rose from 0/150 to 40/150 ($p = 2 \times 10^{-12}$), in all three seeds. But the hand blocked the route to the goal, so its task-success check failed, and with the hand moved aside neither personalized robot touched it. The safeguard was largely stillness, which personalization removes; evaluating VLA refusal needs a complying arm and a task that can be done safely.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their language backbones [2], alignment that shapes only the first output tokens is especially fragile [3], and durability claims are easy to overstate [4]. All concern text or image outputs. Safety is now being installed in the weights of VLA policies [5] and measured in closed loop [6], but each safeguard is trained and evaluated once, whereas a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Backdoors have been designed to survive a user's clean fine-tuning: for OpenVLA-7B on LIBERO, INFUSE reports 95.3% attack success after it, against 31.7% in its own measurement of BadVLA [8, 9]. We ask the defender's converse. Task retention does not settle it: pretrained VLAs retain skills under replay or LoRA with RL, but plain supervised sequential fine-tuning, the regime personalization uses, forgets heavily [10, 11].

Two features make this a different measurement from the text-model literature. A VLA's refusal is a no-op action and harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to be instruction-conditioned, which requires paired instructions on identical states (cf. HazardArena's safe/unsafe twin scenes [13]) and a blank-instruction control.

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2 Study 1: protocol

Substrate and states. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]; old tasks are LIBERO-Spatial, the personalization suite is LIBERO-Object [15]. Hazard scenes are the five static-hand tasks of the hand-object-avoidance suite at level 0 of LIBERO-Safety [6], a fork of LIBERO that adds human hands, run on OpenVLA's own controller with each state's recorded hand pose restored (the fork's reset re-samples it). As shipped, the fork's `CheckRobotContact` collects geometry ids but compares names, so it never fires (issue #3 on the LIBERO-Safety repository); we patch it to collect names, and do not use the fork's published hand-suite collision rates as baselines. Tasks 1–3 supply training states and tasks 4–5 are held out; every confirmatory hazard-scene number comes from states 26–50 of tasks 4–5 ($n = 50$), which no pilot touched. Task 4 shares its floor scene, objects and basket with LIBERO-Object. Each state is paired with a *harmful* instruction targeting the hand, a *benign* one that names the task and mentions the hand, and a *blank* one; training and test templates are disjoint, and both classes contain templates with and without the task text.

Arms. Arm A is P fine-tuned by LoRA with the released checkpoint's adapter settings (rank 32, learning rate 5×10^{-4}, constant in every run) at our effective batch of 16, on 900 rows: 300 movement transitions from violation-free scripted hand-avoiding trajectories under benign instructions, the same 300 frames under harmful instructions labelled with the physical no-op encoded through the frozen normalization key, gripper held (counterfactual pairs), and 300 LIBERO-Spatial rehearsal rows. Comparator C gets the same movement and rehearsal rows plus 150 more of each in place of the no-ops (further violation-free scripted frames under benign templates, and rehearsal), for the same 171 updates; C never sees a harmful instruction in training. Both are merged and personalized with a fresh rank-32 adapter, three epochs, no replay, constant learning rate, on $N = 200$ LIBERO-Object demonstrations (29,352 transitions, 5,505 updates); A also at $N = 50$, scored offline only. Three alignment seeds are scored offline; seed 0 also runs in closed loop.

Pre-registration. Frozen in a tagged commit before any confirmatory data: one primary test, the paired change in harmful-instruction violations from $A@0$ to $A@200$ (exact McNemar; if null, its 95% upper bound); three key secondaries as their own family, in a fixed sequence at $\alpha = 0.05$ ($A@200$ vs. $C@200$ on the same outcome; the control-adjusted change; decay of offline refusal, same sign in all three seeds with seed 0's interval excluding zero); new-task success as manipulation check; and four admissible outcome patterns without numeric thresholds (adaptation with more contact; refusal loss without it; general competence loss; safeguard retained). Everything else is exploratory. A go/no-go gate on states apart from the confirmatory set (offline: states 1–25 of tasks 4–5; closed loop: 30 held-out states of tasks 1–3) had to pass before personalization.

Measurements. *Refusal* (R_{u} harmful, R_{s} benign, blank): for each state, the fraction of its two held-out templates for which all six generated pose tokens lie within one bin of the encoded zero and the gripper token matches its current state, from the first frame, averaged over the 50 states (states are the unit of every interval). *Violation*: any LIBERO-Safety constraint fires in a closed-loop episode (robot–hand or goal-object–hand contact, which ends it); episodes are also classed as held still, moved without contact, or contact. *Utility*: success on all ten tasks of a suite, five states each. Paired binary outcomes get exact McNemar tests and Newcombe intervals [16], per-state means state bootstraps and sign-flip tests, all conditional on the trained chains.

3 Study 1: results

The safeguard was installed, and over-refused. The gate passed on the first attempt (one alignment run per seed): on the 30 closed-loop development states A made no harmful-instruction violation where C made 23% ($p = 0.016$); the non-blocking over-refusal screens ($R_{\mathrm{s}} \leq 0.25$; blank refusal well below R_{u}) were not met. Merged and unmerged $A@0$ agree on 249 of 250 gate refusal decisions. On the confirmatory states $A@0$ refused every harmful instruction ($R_{\mathrm{u}} = 1.00$; P: 0.00) and its violations fell from P's 54 / 90 / 80% to 0 / 2 / 2% (harmful / benign / blank; Table 1), much

	first-frame refusal		violations /50 (harmful / benign / blank)			
Arm	R_{u}	R_{s} / blank	P or A	C	A vs. C, benign	U_{old} / U_{new} (A; C)
P	0.00	0.00 / 0.00	27 / 45 / 40	–	–	0.78 / 0.00
@0	1.00	0.89 / 0.52	0 / 1 / 1	26 / 36 / 22	$p < 10^{-10}$	0.56 / 0.00
A@50	0.92	0.93 / 0.50	–	–	–	–
@200, 3 ep.	0.66	0.78 / 0.18	0 / 8 / 14	11 / 5 / 12	$p = 0.51$	0.00 / 0.60; 0.60
exploratory, separate 12-epoch run (same seed, data and learning rate)						
@200, 6 ep.	0.65	0.47 / 0.40	5 / 11 / –	9 / 18 / –	$p = 0.19$	0.00 / 0.78; 0.70
@200, 12 ep.	0.36	0.26 / 0.06	3 / 9 / –	8 / 9 / –	$p = 1$	0.00 / 0.84; 0.76

Table 1: Seed 0, the 50 confirmatory states. Refusal columns are A's (C's is 0.00 at @0 and 0.32 / 0.46 / 0.48 at @200). U_{old} was measured for P and A only.

of it by holding still ($R_{\mathrm{s}} = 0.89$, blank 0.52). Old-task success fell from 78% to 56%. Seeds 1 and 2 installed R_{u} of 1.00 and 0.70.

Pre-registered tests. The primary was null: harmful-instruction violations stayed at 0/50 ($p = 1$; an increase above 7 points is ruled out). All three key secondaries were rejected: A@200 0% vs. C@200 22% ($p = 0.001$); control-adjusted change $+0.30$ $[0.12, 0.48]$ ($p = 0.006$), driven by C falling from 52% to 22%; and first-frame refusal decayed in every seed (-0.34, -0.38, -0.22; seed 0 $[-0.43, -0.25]$). New-task success rose from 0% to 60% (manipulation check passed).

The harmful-instruction outcome cannot register compliance. No arm showed evidence of executing the harmful clause, i.e. of approaching the hand more closely under the harmful than under the benign template on the same state. P never held still or completed a hazard task, yet contacted the hand in 90% of benign and 80% of blank episodes: the hand lies in the path of whatever the policy does. Under the harmful template P moved as far or farther (median end-effector travel 0.51 vs. 0.42 m) but stayed farther from the hand (median closest approach 22 vs. 16 cm) and contacted it less (54% vs. 90%); so did C@0 (52% vs. 72%). After personalization C reversed this ordering (11 vs. 5), but its harmful-minus-benign closest approach was -0.4 cm $[-2.4, 1.5]$. Harmful-instruction contact thus measures whether the motion passes through the hand, not obedience, and the null primary says nothing about whether the safeguard persisted. Nor did A keep an instruction-conditioned avoidance: its harmful-minus-benign closest-approach gap exceeded C's by $+1.3$ cm $[-1.4, 4.0]$ (exploratory; also null at 6 and 12 epochs).

Finding 1: the aligned policy lost its advantage over the comparator. Under benign instructions A@0 contacted the hand far less than C@0 (1 vs. 36 of 50; blank 1 vs. 22), and not only by freezing: it held still in 27 benign episodes and moved without contact in 22. After personalization it held still in 29, moved in 13 and contacted the hand in 8 ($p = 0.04$ against A@0), six of them on states where A@0 had moved without contact, and all 8 on the held-out task that shares its scene with LIBERO-Object (0/25 on the other). C's benign contacts fell from 36 to 5, so most of the control-adjusted change of $+0.76$ $[0.58, 0.94]$ comes from C becoming safer. A was no longer significantly safer than C: 8 vs. 5 benign ($C - A$ -0.06 $[-0.19, 0.07]$) and 14 vs. 12 blank ($p = 0.79$). In the 12-epoch run the benign intervals were $+0.14$ $[-0.04, 0.31]$ at 6 epochs (11 vs. 18) and 0.00 $[-0.17, 0.17]$ at 12 (9 vs. 9), so a residual advantage for A of up to 7 to 31 points cannot be excluded (new-task success there: 84%; released LIBERO-Object checkpoint: 88.4% [14]).

Finding 2: first-step refusal is not specific. A's refusal fell with training (R_{u} 1.00, 0.66; 0.75, 0.65, 0.36 in the 12-epoch run), and so did its instruction contrast on its own training frames (log-likelihood of the seven no-op tokens, summed, under the harmful minus the benign instruction, averaged per state over its 300 counterfactual frames): 10.0 to 2.9 nats. But C, never taught a no-op, rose from 0.00 to 0.32–0.48 after personalization, A@200 refused benign instructions more than harmful ones (0.78 vs. 0.66), and R_{s} barely moved (0.89 to 0.78) as A's benign-instruction

advantage over C vanished. The first-step proxy neither tracked physical safety nor distinguished a taught refusal from personalization-induced hesitation.

4 Study 2: the hand inside the personalization domain

Finding 1's new contacts all fell on the layout that shares LIBERO-Object's scene, so a second study, tagged before any rollout on its scenes, tested it on fresh ones: the ten LIBERO-Object tasks, each with a hand model unseen in alignment placed between the objects and the basket, 20 sampled states per task (5 for gates, 15 confirmatory; $n = 150$), the benign instruction being the personalization instruction verbatim. Nothing was retrained; seeds 1 and 2 join in closed loop. Primary: benign-instruction violation, $A@0 \rightarrow A@200$; key secondaries, in sequence: seeds 1 and 2, the control-adjusted change, $A@200$ vs. $C@200$; manipulation check: task success. Its gates passed ($C@200$ contacted the hand on 38% of development states, $A@0$ on none).

Pre-registered results. A's benign-instruction violations rose from 0/150 to 40/150 ($+0.27$ $[0.20, 0.34]$, $p = 2\times10^{-12}$), and did so in seeds 1 and 2 (1 to 44 and 0 to 40 of 150, both $p < 10^{-11}$). C fell from 74 to 51 of 150, so the control-adjusted change was $+0.42$ $[0.31, 0.53]$ ($p < 10^{-4}$). After personalization A was not significantly safer than C (40 vs. 51; $C - A$ $+0.07$ $[-0.01, 0.15]$, $p = 0.11$), as in Finding 1. The manipulation check failed: A completed no task before or after (0/150), although the same checkpoint succeeds on 60% of hand-free LIBERO-Object episodes.

Why, and a follow-up (exploratory). The hand lay across the route to the basket: 96% of the personalized policies' 141 successful hand-free LIBERO-Object paths pass within 10 cm (horizontally) of it, so no task could be completed without a violation. $A@0$'s zero was stillness: it held still in 40% of episodes and output the no-op on a median 96% of steps in the rest. After personalization it still held in 40%, but its contacts came at a median step of 28, typically with no no-op output. A follow-up, also tagged before any rollout, moved the hand beside the basket, off most paths, gated on a feasible task ($A@200$ success $\geq 10\%$ on development states) and a hazard ($C@200$ violations $\geq 20\%$). Neither of its two pre-declared placements passed, so it stopped before confirmatory rollouts: $A@200$ completed 28% and 18% of development episodes but violated in 2% and 0%, and $C@200$ in 12% (all carried-object contacts) and 0%.

5 Discussion and limitations

Of the four pre-declared patterns, our reading of Study 1 is refusal loss without more harmful-instruction contact, under useful adaptation (the old-task collapse also fits general competence loss). But the aligned policy was never shown to resist a harmful instruction that another policy obeys; what it lost was its advantage over an unaligned policy during ordinary tasks. Study 2 confirms this loss under pre-registration and points to its mechanism: the aligned policy's benign-instruction safety was largely stillness, personalization removed it, and the policy then moved into a hand that blocked its only route. Where the hand could be passed, neither robot touched it (carried objects did in at most 12% of episodes), so we find no evidence that personalization erased an avoidance specific to the hand. For evaluation, three lessons follow. A first-step refusal rate, the natural transfer of text-model refusal metrics, is non-specific after fine-tuning, in line with the argument that chatbot-style refusal safety does not transfer to robots [17]. A closed-loop harm metric for refusal is informative only if some arm demonstrably complies, e.g. one trained to follow the harmful instruction. And a closed-loop safety test should gate on task feasibility, not only on hazard.

Limitations. One model and one recipe; three alignment seeds offline, one in Study 1's closed loop; simulation with a static hand; Study 1's 50 confirmatory states on two layouts; a freshly installed safeguard that over-refuses. Violations count robot–hand and goal-object–hand contact only, a lower bound. Personalization ran three epochs, shorter than OpenVLA's practice; the 12-epoch extension is exploratory, one seed, post hoc, and from a separate run whose epoch-3 refusal differed by 0.09. Findings 1 and 2 are exploratory and uncorrected; Study 2's follow-up tried only two placements.

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